ORIGINAL RESEARCH

ACUPOINT ELECTRICAL STIMULATION REDUCES ACUTE POSTOPERATIVE PAIN IN SURGICAL PATIENTS WITH PATIENT-CONTROLLED ANALGESIA: A RANDOMIZED CONTROLLED STUDY

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Objectives • The purpose of this study was to evaluate the effect of acupoint electrical stimulation with patient-controlled analgesia (PCA) on reducing acute pain, nausea, and vomiting after surgery for nontraumatic spinal cord injury.

Methods • A randomized, controlled, repeated measures research design was used. Ninety-nine patients undergoing lumbar spinal surgery were randomly assigned to one of three groups. Patients in experimental group 1 (EG1) received true acupoint electrical simulation three times, whereas those in experimental group 2 (EG2) received sham acupoint manually. Patients in the control group (CG) received no acupoint intervention. All patients were measured for pain, initial demand for PCA, demand for opiates, opiate dose, vital signs, and postoperative nausea and vomiting (PONV).

Results • Significant differences were found in postoperative pain, respiratory rate, blood pressure, and opiate doses across time in the three groups with better outcomes observed in EG1 than in EG2. However, no between-group difference was found in initial demand for PCA or in PONV.

Conclusions • The study demonstrates that acupoint electrical stimulation improves acute postoperative pain management without adversely affecting vital signs after surgery for nontraumatic spinal injury. More studies are needed to evaluate the effects of acupoint electrical stimulation on PONV and postoperative pain following other surgical procedures. (*Altern Ther Health Med*. 2010;16(6):10-18.)

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Pain is one of the most common adverse postoperative outcomes. Research has shown that 77% of patients who undergo thoracic, upper abdomen, or orthopedic surgery experience pain, and 80% to 85% of these patients experience moderate-to-severe pain.[1] Operative wounds can be the source of more pain than anticipated, especially in patients undergoing spinal surgery.[2] Unrelieved pain can lead to increased cardiac loading and oxygen consumption, which can lead to increased risk of deep vein thrombosis and consequent pulmonary embolism.[3]

Patient-controlled analgesia (PCA) is a safe, effective, and patient-oriented approach to deal with this problem. The main goal of PCA, apart from continuous drug administration, is to give patients access to their analgesia after each lockout interval has elapsed. When the PCA device is activated by pressing a button, a fixed PCA bolus dose is injected so that the circulating analgesic can be maintained at a stable and effective concentration.[4] PCA can adequately manage acute surgical pain, but giving analgesics on an "as needed," or pro re nata, basis cannot.[5] In one study on PCA after spinal surgery, Gepstein et al found that most patients (213 of 237) were satisfied with the PCA procedure,[6] and in a study by Inoue et al, most surgical nurses agreed that PCA is a good way to achieve postoperative analgesia.[7] However, postoperative nausea and vomiting (PONV) occurs as PCA-related side effects in about 43% of cases, which could be reduced by decreasing opiate doses of PCA.

Surgical patients receiving general anesthesia have a high incidence of PONV.[9] Treating PONV with drugs is routine practice.[10] Although administering droperidol significantly lowers the incidence of PONV, a high dose does not guarantee increased postoperative well-being and may cause side effects (eg, increased restlessness, anxiety, disturbed sleep).[11] The use of antiemetic drugs also reduces the relative risk of PONV by 33.5%.[12] However, taking these drugs may necessitate prolonged bed rest and lead to aspiration pneumonia as well as esophageal rupture, increasing the length of hospital stays and the mortality rate.[13] Alkaissi et al showed that the intensity of symptoms and the total number of distress-inducing symptoms did not decrease after antiemetic treatment in patients at high risk for PONV.[14]

Patients (especially Chinese surgical patients) may assume passive roles in their pain management and therefore receive far less postoperative pain medication than is required.[15,16] For this reason, efforts to improve pain management should move beyond assessing pain to implementing and evaluating improvements in pain intervention.[17] Health care professionals have accepted and adopted acupoint stimulation as one such intervention in the treatment of pain and PONV. The use of acupoint stimulation has 1000 years of history and much evidence for its effectiveness. In addition, it can be administered and effective without piercing the skin.[18] A special electrical waveform applied across the intact surface of the skin to activate nerve fibers involved in analgesic mechanisms may be sufficient for rapid, effective, and long-lasting pain relief.

A meta-analysis of 21 randomized controlled trials demonstrated that applying electrical stimulation reduced analgesic consumption for postoperative pain by 35.5%.[19] In another study, PCA combined with low frequency (2 Hz) and high frequency (100 Hz) electroacupuncture in the first 24 hours after surgery reduced total dose of PCA by 27% and 35.8%, respectively.[20] In a separate study, electroacupuncture postponed the initial demand for pain control and decreased the total PCA dose needed in the first 24 hours after surgery.[21] In patients who had cesarean sections and required PCA, electroacupuncture also postponed the initial demand and reduced the total volume of opiate consumed for pain control in the first 24 hours after surgery.[22]

Some studies found that prevention of PONV simultaneously reduced patients' pain. A meta-analysis conducted by Lee et al found that nonpharmacological techniques, including electroacupuncture, electrical stimulation, acupuncture, and acupressure, were more effective than placebo in preventing PONV in surgical patients.[23] Chen et al found that PCA and electroacupuncture (100 Hz) prior to surgical anesthesia not only reduced the amount of morphine required but also reduced incidence of nausea during the first 24 hours after surgery.[21] Cekmen et al also found that stimulation at a frequency of 5 Hz for 50 milliseconds at a current density of 0.5 mA to 4 mA reduced postoperative analgesia demand, use of antiemetic drugs, and incidence of PONV.[24] However, Zárate et al demonstrated that acupuncture electrical stimulation significantly reduced the incidence and severity of postoperative nausea but not of vomiting or retching.[25] Chen et al demonstrated that opiate requirement and PONV effectively decreased with acupoint electrical stimulation compared with the sham and the control.[26]

A review of the literature showed that acupoint stimulation, such as acupuncture, when used for postoperative pain, has short-term benefits and inconclusive effects, depending on the timing of the intervention and the patient's level of consciousness.[27] Researchers have not conducted in-depth studies using acupoint electrical stimulation either as a sole analgesic or as supplementary analgesia.[28] Table 1 summarizes acupuncture interventions that have worked best under specific conditions. Well-designed, sham-controlled clinical trials need to verify the effects of acupoint stimulation by using clinical outcome measures.

This study evaluates the effect of acupoint electrical stimulation with PCA to reduce acute postoperative pain and PONV after surgery for nontraumatic spinal cord injury. We investigated the following hypotheses:

1. Significant differences exist in postoperative pain among the three groups.
2. Significant differences exist in vital signs among the three groups.
3. Significant differences exist in PONV among the three groups.

MATERIALS AND METHODS

The present study used a randomized-controlled, repeated-measures research design (Figure 1). Patients were randomly assigned to one of three groups. Experimental group 1 (EG1) received true acupoint intervention via electrical stimulation, experimental group 2 (EG2) received the sham acupoint manually, and the control group (CG) received no acupoint stimulation. Only one of the researchers knew the randomization result. Patients and their nurses and physicians were not informed of the group allocation. Data were collected after the intervention three times on the first operative day. Pain intensity, vital signs, initial demand for PCA, demand for opiates, opiate dose, and incidence of PONV were evaluated. All patients were scheduled for surgery to correct nontraumatic lumbar spine injuries under general anesthesia and had been evaluated as American Society of Anesthesiologists grades I to II (I: a normal healthy patient; II: a patient with mild systemic disease that does not limit activity). All had consented to an epidural PCA with a 1-mg bolus of morphine and a 10-minute lockout interval without basal infusion. The exclusion criteria included the presence of a pacemaker; a history of epilepsy or opiate dependence; and skin infection, damage, or hemorrhage at the site of acupoint application. The use of other treatments for pain and PONV during the study period was also excluded. With a medium effect size of $f=0.25$,[29] significance level of $\alpha = 0.05$, repetition of 3, correlation among repetition measures of 0.3, and statistical power of $1-\beta = 0.8$, the sample size of 87 was estimated from visual analogue scale (VAS) scores.

Interventions

Electrical stimulation was applied to four pairs of acupoints, Weizhong (BL40), Yanglingquan (GB34), Shenmen (HT7), and Neiguan (P6), which reduce pain[18,21,30] and PONV.[18,31] According to traditional Chinese medicine (TCM), electrical stimulation on acupoints is the same as acupuncture; the continuous flow of *qi* (life energy) and blood is vital for improving

TABLE 1 Key Studies of Acupoint Stimulation*

First Author	Surgery (N)	Intervention	Acupoint	Control	Primary Outcome	Results
Wei 2001[20]	Cesarean section (121)	• PCA+low frequency EA (2 Hz, 30 min) • PCA+high frequency EA (100 Hz, 30 min)	SP6	Sham EA Standard control	• Pain intensity • Opioid requirement	EA better than control group and sham EA for all measured parameters for the first 24 h
Chen 2004[21]	Arthroscopic surgery (60)	• PCA+EA (100 Hz, 30 min before anesthesia)	GB34, SP9	Sham EA Standard control	• Opioid requirement	During the first 24 h, analgesic requirement decreased in EA vs sham or control
Lin 2004[22]	Cesarean section (90)	• PCA+EA (100 Hz, 30 min before anesthesia) • PCA+acupuncture	SP6	Standard control	• Pain intensity • Opioid requirement • Opioid-related side effect	EA better than control group and sham EA for all measured parameters for the first 24 h
Cekmen 2007[24]	Laparoscopic cholecystectomy (40)	• TENS (5 Hz, 50 ms, 0.5-4 mA for 6 h)	Trapezoid area	Standard control	• Nausea and vomiting • Analgesic and antiemetic need	TENS better than control for all measured parameters
Zárate 2001[25]	Laparoscopic cholecystectomy (221)	• TAES (Relief Band used for 9 h after surgery)	P6	Sham Placebo	• Nausea and vomiting	TAES with the Relief Band at P6 reduced nausea but not vomiting
Chen 1998[26]	Abdominal hysterectomy or myomectomy (100)	• PCA+AES (2/100 Hz per 3 s, 9-12 mA, 30 min, bilateral, use TENS per 2-3 h while awake)	ST36	Sham Non-AES ES	• Opioid requirement	TENS applied at dermatome level of the skin incision is as effective as Zusanli acupoint stimulation
Wang 1997[38]	Abdominal surgery (76)	• PCA+low-TAES (4-5 mA,) • PCA+high-TAES (9-12 mA, alternating 2/100 Hz per 3 s, use TAES for 30 min per 2 h while awake)	Di4	PCA only Sham-TAES	• Pain intensity • Opioid requirement • Opioid-related side effect • Requirement for antiemetic	High-TAES produced a significant decrease in the PCA opioid requirement and the opioid-related side effects after lower intra-abdominal surgery
Chao 2007[39]	Labor (100)	• TAES (10-18 mA, 2/10 0 Hz for 30 min, bilateral, pulse duration 0.25 ms)	Li4, SP6	Placebo	• Pain intensity • Satisfaction of pain relief and willingness to repeat treatment	TAES better than the placebo group for all measured parameters in the first lab stage
Gupta 1999[40]	Knee arthroscopy (42)	• Acupuncture (ipsilateral to the side of surgery, needles remained for 15 min and manually stimulated for 5 s per 5 min and just before needle removal)	SP9, SP10, ST34, ST36, LI4	Standard control	• Pain intensity • Opioid requirement	No significant difference between the groups in any outcome
Hamza 1999[41]	Gynecological surgery (100)	• PCA+TENS (2 Hz) • PCA+TENS (100 Hz) • PCA+TENS (alternating 2/100 Hz per 3 s)	Next to skin incision	PCA+ sham TENS	• VAS for pain, sedation, fatigue, and nausea • Opioid requirement	Mixed frequency of stimulation decreased morphine requirements compared with the sham and low- and high-frequency groups
Kober 2002[43]	Minor trauma (60)	• True acupressure (no reported time)	Di4, KS9, KS6, BL60, LG20	Sham acupressure Standard control	• VAS for pain intensity, anxiety, and satisfaction	At the end of transport true points group better than the control and sham for all measured parameters

*PCA indicates patient-controlled analgesia; EA, electroacupuncture; AES, acupoint electrical stimulation; TAES, transcutaneous acupoint electrical stimulation; TENS, transcutaneous electrical nerve stimulation; ES, electrical stimulation; VAS, visual analogue scale.

Groups	Before Surgery	After Surgery I	After Surgery II
EG1	O_1X_1	$O_2X_1O_3$	$O_4X_1O_5$
EG2	O_1X_2	$O_2X_2O_3$	$O_4X_2O_5$
CG	O_1	O_2 O_3	O_4 O_5

FIGURE 1 Research Design*

*Before surgery indicates 1 hour before surgery; After surgery I, 1 hour after returning to the ward from the operating room; After surgery II, 2 hours after returning to the ward from the operating room; EG1, experimental group 1; EG2, experimental group 2; CG, control group; X_1, true acupoint stimulation; X_2, sham acupoint stimulation; O_1, Brief Pain Inventory (BPI), Rhodes index of Nausea and Vomiting (RINV), and vital signs; $O_{2\text{-}3}$ BPI and vital signs; $O_{4\text{-}5}$, BPI, RINV, and vital signs.

one's health by electrically stimulating specific acupoints on the body that are thought to rectify *qi* and blood.[18] In this study, two pairs of disposable rubber electrode pads (30 mm x 30 mm) were placed bilaterally at the four points, and electrical stimulation was applied with a frequency of 100 Hz, a brush frequency of 2 Hz (dense-dispersed waveform), and a pulse duration of 0.25 milliseconds, for a total of 20 minutes at three different times (1 hour before surgery, and 1 hour and 2 hours after the patient returned from the operating room).[21,22,30-32] The current output was titrated for each individual with an intensity in the range of 4 mA to 7 mA depending on body weight. Patients were supine throughout the procedure. The same procedure was applied to patients in EG2 but at different (sham) acupoints that were 2 cm apart from a true acupoint and not on any meridian.[30] The researcher who administered the intervention had previously earned 40 TCM-related formal credits and was qualified by three acupuncture doctors to perform the intervention. Content validity of the acupoint electrical stimulation protocol was verified by two physicians practicing TCM. Validity was established if the acupoint selection and location, current frequency, current output, waveform, length of stimulation, and efficacy were appropriate and if the content validity index (CVI) was 1.0. To make certain of the neurophysiological, neurochemical, and psychological impacts of the procedure, we included the CG group in repeated measures analysis, but these patients did not receive any electrical stimulation.

Measurements

In this study, we measured and observed the primary outcomes of pain and the secondary outcomes of vital signs and PONV. Pain was measured using the severity and interference with daily living subscales of the Brief Pain Inventory (BPI).[33] A VAS was used to evaluate pain intensity.[34] On the scale, 0 represented no pain at all; 1 to 3, mild pain; 4 to 6, moderate pain; 7 to 9, severe pain; and 10, intolerable pain. Because patients perform daily activities on the first postoperative day with difficulty, work-related items on the interference subscale were replaced by deep breathing and coughing, and daily activities included eating, dressing, and personal toilet. Cronbach's α was 0.65 for pain severity and 0.78 for pain interference in this study.

PONV was measured with the Chinese version of the Rhodes Index of Nausea and Vomiting (RINV).[35] This index, an 8-item questionnaire with a 5-point Likert-type (0 to 4) response format, elicits information about experience, occurrence, duration, and distress associated with nausea, vomiting, and dry heaves. A higher score indicates more severe PONV. Cronbach's α ranged from 0.89 to 0.97,[36] and overall it was 0.82 in this study. In addition to questionnaires, we used the Sure Signs (C1) device made by Philips Company (Bothell, Washington) to measure vital signs, including respiratory rate, heart rate, and blood pressure. The system used for PCA (IL60064) was made by Abbott Laboratories (North Chicago, Illinois).

Demographic characteristics included age, gender, marital status, employment, body weight, and body height. Disease characteristics included disease history, preoperative pain, type of surgery, surgery duration, and amount of blood loss. Vital signs of heart rate, respiration rate, and blood pressure were measured. Pain control was evaluated by means of a questionnaire containing items for dose, medication, patient self-assessment, side effects, and vital signs, with a CVI of 0.89 and Cronbach's α of 0.93.[37]

Data Collection and Analysis

This study was carried out by the orthopedic departments of a 4000-bed medical center in northern Taiwan. Ethical approval was obtained from the institutional review board of the same hospital. Patients who met the criteria for inclusion were recruited, and informed consent was obtained from all participants. The purpose and procedure of the study were explained to the participants, who were informed that data collected would remain confidential and their anonymity would be protected. All participants were free to withdraw from the study at any time. Data of primary and secondary outcomes and demographic factors were collected immediately after the intervention.

Data were analyzed with SPSS 17.0 for Windows (SPSS Inc, Chicago, Illinois). Demographic data and medical history data were analyzed with the use of descriptive statistics and the chi-square test to verify homogeneity among the groups. The effects of acupoint electrical stimulation among the groups were analyzed with a 1-way and repeated-measures analysis of variance (ANOVA) with the Sheffe post hoc test.

RESULTS

Initially, 99 participants met the criteria and were included in the study. Five participants did not complete the study due to unstable conditions. Of the remaining 94 participants, 33 were in the EG1, 30 in the EG2, and 31 in the CG. The attrition rate was 5%. Patient ages ranged from 23 to 88 years with a mean of 60.4 (SD = 13.9); 38.3% (n = 36) of patients were male, and 61.7% (n = 58) were female. In addition, 84.0% of patients (n = 79) were married, and 72.3% of subjects (n = 68) were unemployed. Table 2 summarizes the demographic and disease characteristics of each group. All characteristics except marital status were similar in the three groups.

TABLE 2 Homogeneity of the Demographic and Disease Characteristics Among the Groups*

Variables	EG1 (n=33)	EG2 (n=30)	CG (n=31)	F or χ^2 (P)
Gender (n/%)				
Male	11/33.3	9/30.0	16/51.6	χ^2=3.54 (.17)
Female	22/66.7	21/70.0	15/48.4	
Age (M±SD)	60.7±12.0	57.2±15.7	63.2±14.0	F=1.43 (.24)
Marital status (n/%)				
Married	30/90.9	21/70.0	28/90.3	χ^2=6.48 (.04)
Other	3/9.1	9 /30.0	3/9.7	
Employment (n/%)				
Employed	11/33.3	8/26.7	7/22.6	χ^2=0.95 (.62)
Unemployed	22/66.7	22/73.3	24/77.4	
Body height, cm (M±SD)	158.1±7.7	157.1±9.9	160.5±9.0	F=1.18 (.31)
Body weight, kg (M±SD)	65.3±10.1	63.9±13.6	64.7±13.1	F=0.09 (.91)
Numbers of chronic disease history (n/%)				
None	20/60.6	17/56.7	11/35.5	
1	6/18.2	8/26.7	9/29.0	χ^2=10.27 (.11)
2	6/18.2	5/16.6	6/19.4	
3	1/3.0	0	5/16.1	
Diagnosis				
HIVD	7/21.2	5/16.7	2/6.5	
Lumbar vertebra dislocation	13/39.4	10/33.3	6/19.4	χ^2=8.32 (.08)
Vertebral foramen stenosis	13/39.4	15/50	23/74.2	
Experience of lumbar surgery (M±SD)	0.4±0.9	0.3±0.8	0.3±0.5	F=0.28 (.76)
Worst preoperative pain (M±SD)	7.1±1.9	7.2±1.1	7.16±1.7	F=0.03 (.97)
Average preoperative pain (M±SD)	1.5±1.1	1.3±1.0	0.9±0.9	F=2.95 (.06)
Preoperative respiratory rate (M±SD)	18.7±1.2	18.4±1.1	18.7±1.1	F=0.72 (.49)
Preoperative heart rate (M±SD)	71.2±9.6	70.1±8.1	65.9±11.4	F=2.49 (.09)
Preoperative systolic blood pressure (M±SD)	131.3±20.2	127.3±17.4	137.9±18.7	F=2.49 (.09)
Preoperative diastolic blood pressure (M±SD)	75.9±9.8	74.5±10.6	80.6±11.5	F=2.75 (.07)
Type of operation (n/%)				
Lumbar discectomy	27/81.8	25/83.3	24/77.4	
Spinal decompression	21/63.6	24/80.0	28/90.3	χ^2=6.00 (.20)
Spongy bone transplantation	16/48.5	23/76.7	22/71.0	
Lumbar laminectomy	2/6.1	0/0.0	3/9.7	
Operation duration (min) (M±SD)	216.4±93.9	261.3±118.4	270.6±124.5	F=2.13 (.13)
Amount of blood loss (mL) (M±SD)	456.8±436.8	477.7±459.1	453.6±351.2	F=0.03 (.97)

*EG1 indicates experimental group 1; EG2, experimental group 2; CG, control group; HIVD, herniated intervertebral disc.

Pain Assessment

Figure 2 shows the pain trend over time for each group. Table 3 shows the pain assessment results by each group and the repeated ANOVA measures with the Scheffe hoc test. Significant differences were observed in pain intensity (F=6.36, P<.001), and average pain (F=5.22, P=.01) among the groups, with the worst pain experienced in the first postoperative 24 hours (F= 5.83, P<.001). Significant differences were also found in these results between EG1 and EG2 (P=.02, .02, .03, respectively) and between the EG1 and the CG (P=.01, .02, .02, respectively). In addition, the pain score was significantly different among the three groups (F=5.74, P<.001) and between EG1 and the CG (P =.01) when patients carried out routine activities (eg, changing position, deep breathing, and coughing). Significant differences were also found in the impact of postoperative pain on sleep (F= 3.05, P=.05), without a between-group difference on the Scheffe post hoc test.

Vital Signs Assessment

Figure 3 shows the changes over time in vital signs (respiratory rate, heart rate, and blood pressure) in each group and the repeated ANOVA measures with the Scheffe post hoc test of these data. Significant differences occurred in respiratory rates ($F=1.90$, $P<.001$), systolic pressure ($F=8.03$, $P<.001$), and diastolic pressure ($F=8.89$, $P<.001$) among the groups, and between the CG and EG1 ($P<.001$, .001, .001, respectively) as well as between the CG and EG2 ($P<.001$, .001, .001, respectively). In contrast, heart rates were similar among the groups ($F=1.22$, $P=.33$).

Opiate Demand and Dose

Table 4 shows the results of ANOVA with the Scheffe post hoc test for opiate demand and dose in the first postoperative 24 hours. Findings demonstrated no significant difference in the initial demand for PCA among the three groups ($F=0.13$, $P=.88$). However, the number of demands ($F=5.93$, $P<.001$), and the total dose ($F=4.28$, $P=.02$) were significantly different among the 3 groups and between the EG1 and CG ($P=.02$).

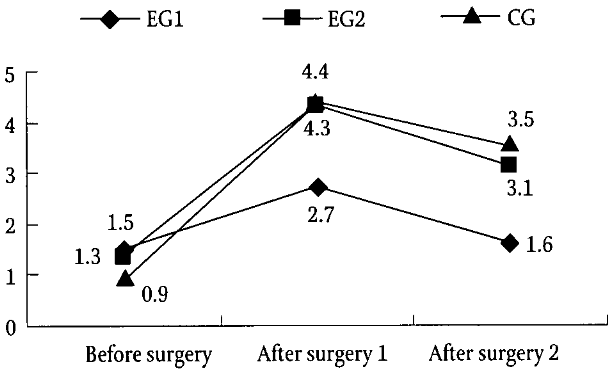


FIGURE 2 Pain Intensity Changes Across Time

*$F=6.36$ ($P<.001$), †EG2 > EG1, G > EG1. Before surgery indicates 1 hour before surgery; After surgery I, 1 hour after returning to the ward from the operating room; After surgery II, 2 hours after returning to the ward from the operating room. †Scheffe post hoc test.

Postoperative Nausea and Vomiting

Table 5 shows the results of ANOVA with the Scheffe post hoc test for PONV. Findings showed no significant difference overall ($F=-0.97$, $P=.33$) or in any item of the PONV among the groups ($P>.05$).

DISCUSSION

This study demonstrated the effect of acupoint electrical stimulation on improving acute postoperative pain, pain impact, opiate demand, and PONV in postoperative patients who received PCA. All demographic and disease characteristics except marital status were similar among the three randomized groups. Baseline severities of the worst and average preoperative pain experienced were also similar among the groups.

After intervention, acupoint electrical stimulation reduced pain but not PONV. Patients in the true acupoint electrical stimulation group experienced less pain severity, pain frequency, and pain impact on daily activities and sleep in the first postoperative 24 hours than did patients in the other groups. This result is consistent with pain reduction findings in other studies.[20-22,37] These studies validated the effect of acupoint electrical stimulation in conjunction with PCA in relieving the operative pain[20] and postoperative pain[22] in patients who had cesarean sections and in relieving the postoperative pain in patients who had total knee replacements.[21,37] The results of this study are also similar to those of other studies[38,39] that examined the effects of acupoint electrical stimulation without PCA on the postoperative pain and

TABLE 3 Comparison of Pain Among Groups

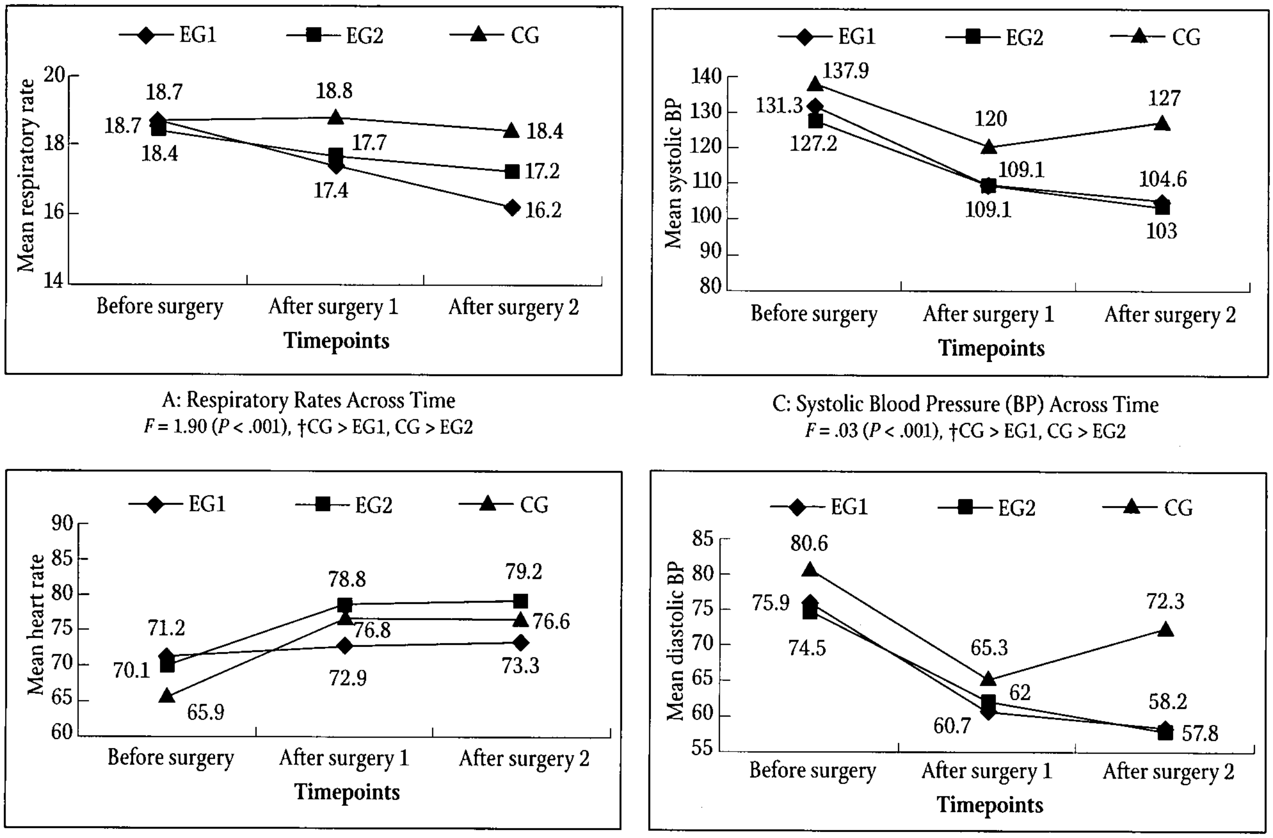
Variables	EG1 (n = 33) M ± SD	EG2 (n = 30) M ± SD	CG (n = 31) M ± SD	Range	F (P)
Worst postoperative pain	5.3±1.5	6.5±1.6	6.5±1.6	2-10	F=5.83 (<.001) †EG2>EG1, G>EG1
Average postoperative pain	2.1±1.2	3.0±1.4	3.1±1.5	0-7	F=5.22 (.01) †EG2>EG1, G>EG1
Impact of postoperative pain on daily activities	5.3±1.4	6.1±1.5	6.5±1.7	2-10	F=5.74 (<.001) †CG>EG1
Impact of postoperative pain on sleep	1.3±1.5	2.1±2.2	2.3±1.6	0-7	F=3.05 (.05) †nil

*Pain intensity scores on the visual analogue scale. EG1 indicates experimental group 1; EG2, experimental group 2; CG, control group. †Scheffe post hoc test.

TABLE 4 Comparisons of Opiate Demands and Dose of PCA Among Groups*

Variables	EG1 (n = 33) M ± SD	EG2 (n = 30) M ± SD	CG (n = 31) M ± SD	F (P)
Initial demand launched (min)	98.4±156.2	92.8±109.1	81.5±131.7	F=0.13 (.88)
Occurrence of PCA button pushed (time)	25.3±28.5	47.4±39.7	74.7±85.5	F=5.93 (<.001) †CG>EG1
Dose of opiate used in 24 h (mg)	18.6±9.7	21.6±13.1	27.2±12.5	F=4.28 (.02) †CG>EG1

*PCA indicates patient-controlled analgesia; EG1, experimental group 1; EG2, experimental group 2; CG, control group. †Scheffe post hoc test.



A: Respiratory Rates Across Time
$F = 1.90$ ($P < .001$), †CG > EG1, CG > EG2

C: Systolic Blood Pressure (BP) Across Time
$F = .03$ ($P < .001$), †CG > EG1, CG > EG2

B: Heart Rate Across Time
$F = .22$ ($P = .33$)

C: Diastolic Blood Pressure (BP) Across Time
$F = 8.89$ ($P < .001$), †CG > EG1, CG > EG2

FIGURE 3 Vital Sign Changes Across Time*

*Before surgery indicates 1 hour before surgery; After surgery I, 1 hour after returning to the ward from the operating room; After surgery II, 2 hours after returning to the ward from the operating room. †Scheffe post hoc test.

opioid analgesic requirements in patients who had lower abdominal surgery or were in the first stage of labor.[39] Regardless of the different meridian points used in different types of surgery, electrical stimulation on the appropriate corresponding points eases pain and reduces analgesia requirements. According to TCM, acupoint stimuli (transmitted to the brain and other organs via nerves and meridian lines) modulate physiological reactions.[18] In the group receiving acupoint stimulation in the present study, pain intensity decreased over time. Similarly, another study demonstrated that electroacupuncture relieved pain in a rapid, efficient, and long-lasting manner.[21] However, in contrast to these studies, one sham-controlled study on the effect of acupuncture did not demonstrate a decrease in the analgesic requirement after knee arthroscopy.[40] In addition, no evidence-based proof has been found for the effect of acupuncture beyond the 24-hour postoperative period.

In the present study, acupoint electrical stimulation 1 hour before surgery did not delay the time to launch PCA, but it did decrease the number of demands for PCA and the total amount of PCA needed after surgery. This result contrasts with findings from other studies[20-22] in which electroacupuncture with PCA postponed the initial demand for pain control in the first postoperative 24 hours. Even though stimulation in our study did not significantly delay the initial demand for pain control, the clinical significance of our results notably echoes that of other studies in which a delay was found. In general, more invasive and postoperative recovery is longer for spinal surgery than cesarean section[20,22] or knee joint surgery.[21] This longer recovery time may reduce the effect of electroacupuncture on analgesia at the time of initial demand. Nevertheless, in our study, stimulation reduced the total amount of opiate used in the first postoperative 24 hours. This observation is in agreement with that of other studies.[20-22] Moreover, one study found that electroacupuncture stimulus reduced the overall dose of opiate demanded and consumed in the first 24-hour postoperative period.[41] Thus, further clarification of pain relief at initial demand, prolongation of the effect, and initiation and duration of stimulation is required.

In our study, the worst postoperative pain (experienced by patients while performing routine activities) was less severe in the group that received true acupoint electrical stimulation than it was in the other groups. Notably, the patients in the true stimulation group stated that their level of pain was moderate. Fewer opiates were needed to achieve pain relief in the stimulation group, even though all

TABLE 5 Comparisons of Postoperative Nausea and Vomiting Among Groups*

Variables	EG1 (n = 33) M ± SD	EG2 (n = 30) M ± SD	CG (n = 31) M ± SD	Range	*F (P)*
Frequency of nausea	0.3±1.1	0.5±1.3	0.8±1.4	0-6	*F*=1.59(.21)
Distress from nausea	0.3±1.1	0.5±1.7	1.2±2.0	0-8	*F*=2.06 (.08)
Duration of nausea	0.1±0.3	0.2 ±0.8	0.4±0.6	0-4	*F*=2.17 (.12)
Frequency of vomiting	0.2±0.6	0.4±1.3	0.1±0.3	0-5	*F*=1.62 (.20)
Amount of vomiting	0.1±0.4	0.3±0.9	0.1±0.6	0-4	*F*=0.95 (.39)
Distress from vomiting	0.2±1.0	0.5±1.6	0.3±1.0	0-8	*F*=0.41 (.66)
Frequency of dry heaves	0.0±0.0	0.1±0.7	0.1±0.4	0-4	*F*=0.68 (.51)
Distress from dry heaves	0.0 ±0.0	0.2±0.9	0.2±0.9	0-5	*F*=0.84 (.43)

*EG1 indicates experimental group 1; EG2, experimental group 2; CG, control group.

groups used less than the expected dose of opiate. This is in agreement with studies[16-17,37] that showed that Chinese patients ask for and receive very small amounts of postoperative pain medication. Most Chinese patients are concerned about drug addiction[24] and social stigma.[17] The opiate dose used in the first postoperative 24 hours was higher in one study[41] than in the present study. However, more research is required to address matters of opiate dose.

Severe postoperative pain can destabilize the heart rate and blood pressure, depress respiration, and limit a patient's activities. Pain can be released by opiates that have a direct effect on the respiratory center in the pontine and bulbar brain stem. Respiratory depression often occurs as a result of excessive doses or frequent administration of opiates.[42] In the present study, however, vital signs, (heart rate, respiration, and blood pressure) were normalized. Furthermore, the worst postoperative pain occurred while patients were doing daily activities and it influenced vital signs. The use of PCA provides pain control with a low risk of respiratory depression.[42]

In the present study, acupoint electrical stimulation reduced patients' postoperative pain and opiate requirements without adverse effects on vital signs. Similarly, one study found that stimulating certain acupoints resulted in significantly reduced pain intensity and heart rate.[43] Another study found that acupuncture analgesia stabilized blood pressure, respiration, and heart rate during surgery.[44] Acupoint stimuli thus elicit impulses or volleys that travel via small myelinated nerve fibers to the spinal cord, midbrain, pituitary gland, and hypothalamus.[45] Clinical observation in the present study demonstrated that patients in the true acupoint group had less pain and opiate demand and had more tolerance for activities than did patients in other groups. The conclusive findings of this study, that acupoint stimulation reduces postoperative pain and stabilizes respiration and blood pressure but not heart rate, echo those of the previously mentioned studies.

Our study did not find that stimulation reduced the incidence of nausea, vomiting, and dry heaves. This result is in agreement with the findings of Agarwal et al that 30 minutes of acupoint stimulation by wrist laces does not reduce PONV[46] and those of Gan et al that 30 minutes of acupoint electrical stimulation is ineffective in reducing the incidence of vomiting.[47] In contrast, a review by Lee et al[23] as well as other studies[21,24] found that acupoint stimulation is more effective in preventing PONV than is placebo for surgical patients. Dundee et al also showed that 5 minutes of acupoint electrical stimulation with a low-frequency current markedly reduces PONV,[48] and Ming et al showed that 5 minutes of acupressure on each point 1 hour before surgery prevents PONV.[31] However, Zárate et al showed that acupoint electrical stimulation reduces nausea but not vomiting.[25] In our study, 90.9% of the acupoint electrical stimulation group had few or no symptoms of PONV and few occurrences of PONV. This result may be because the dose of morphine used in the first postoperative 24 hours in this study was lower, compared with that in the study by Aubrun et al in which 33 mg and 49 mg of morphine were administered with vomiting rates of 28% and 32% in the experimental and control groups, respectively.[49] A low vomiting rate could be explained by the corresponding low PCA usage in our study if morphine dosage is the most predictive PONV factor.[50] Further studies should verify these clinical effects.

CONCLUSION

Achieving postoperative pain relief is a complex problem. Because this study was conducted in one hospital, the findings may not be representative and may not be generalized to other patient groups. However, by using a randomized-controlled, repeated-measures design, we demonstrated that true acupoint electrical stimulation results in significant acute pain reduction, pain impact, and opiate demand, as well as stabilized respiration and blood pressure in the first 24 hours after lumbar spinal surgery. Findings did not include a reduction in PONV. The acupoint electrical stimulation technique is economical and easy and safe to use. Studies that integrate Chinese medicine with Western medicine for managing operative pain and PONV can provide valuable references for clinical practice.

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